How Much Does Body Pose Help Pedestrian Trajectory Forecasting? A Seed-Replicated Study and Auxiliary-Loss Audit on Waymo

Anonymous authors
Paper under double-blind review

Abstract

Body pose is a plausible but rarely used signal for pedestrian trajectory forecasting in autonomous driving. We extend the Motion Transformer (MTR) with a GRU encoder over past SMPL body poses (6D rotation representation) and measure what pose conditioning is actually worth on a pedestrian subset of the Waymo Open Motion Dataset, using LiDAR-derived pose pseudo-ground-truth. Our study has three parts. First, a *data-forensics audit*. An earlier draft of this work reported that pose helps in one configuration. We then found that the SMPL export keyed pose clips by a scene-*local* object id across all $\sim$326k scenes, so nearly every training sequence carried *another pedestrian's* body motion and $4.7\times$ of the pedestrians were silently dropped; the correct-pose condition had never been run. A per-scene re-export (299,220 scenes, 1.81M sequences) cut root-heading error from 41.5° to 11.4° and raised gait–speed correlation from $r{=}0.26$ to 0.58.

Second, a *seed-replicated measurement on the corrected data*, in which the decisive comparison is never pose against no-pose—which confounds information with branch capacity—but correct pose against the *same architecture fed another pedestrian's pose*. On that test we find nothing, anywhere: from scratch -0.66% ($p{=}0.46$), fine-tuned -0.88% ($p{=}0.11$), and at $40\times$ more data -0.04% ($p{=}0.97$), while the baseline itself improves 19% from data alone. A directed agentic architecture search over the pose-integration code (two rounds, 20 scored candidates) accepted none, and its most promising candidate, which looked significant on a truncated proxy, measured -0.04% ($p{=}0.92$) at $n{=}8$ seeds under the full recipe on a held-out split.

Third, a *mechanism*. Body facing is genuinely informative for pedestrians who are standing still and then step off—but the tracker's box heading already estimates it, more accurately than mesh fitting recovers it (25.2° vs. 44.2° median error), and that channel is already an input to the model; pose adds nothing on top of it in any range, occlusion or jitter stratum. Per-joint scoring against ground-truth keypoints shows why the head in particular is unusable: it carries a tenth of the skill of any limb joint while its MPJPE looks better than either wrist. We report ten self-caught artifacts, including three of our own headline claims, and conclude with the condition under which an auxiliary modality can pay at all.

1 Introduction

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry—but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner's body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023). While recent work uses skeleton keypoints for pose-conditioned

prediction in robot navigation and pedestrian benchmarks (Salzmann et al., 2023; Saadatnejad et al., 2024; Gao et al., 2025), state-of-the-art autonomous driving forecasters operating on Waymo (Ettinger et al., 2021) discard body pose entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction within a SOTA AV forecasting framework, and if so, what design choices drive the gain?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and auxiliary future-pose prediction heads. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with LiDAR-derived SMPL pose pseudo-ground-truth from Waymo-3DSkelMo (Zhu et al., 2025), using minADE and minFDE over 6 predicted modes.

This paper deliberately reports both a *result* and a sequence of *corrections of our own earlier results*, because the corrections are the instructive part. An earlier single-run draft credited geodesic SO(3) supervision with -7.6% minADE; an audit showed that was impossible—the future pose targets are absent, so the loss is a constant with identically zero gradient and the two runs compared were the same experiment under different seeds (Section 4.8). A later draft, rebuilt with five seeds, reported that pose helps in one configuration. That claim did not survive either: its own shuffled-pose control reproduced the effect, and we then discovered that the pose export had been keyed by a scene-local object id, so every pose-conditioned run we had ever performed trained on another pedestrian's body motion. Everything below is measured on a corrected per-scene re-export.

- **No pose benefit, at any scale or regime, against the control that matters.** The decisive comparison is not pose versus no-pose—which confounds information with the capacity of an extra input branch—but correct pose versus the *same architecture fed a different pedestrian's pose*. That test is null from scratch (-0.66%, p=0.46), after full-Waymo pretraining (-0.88%, p=0.11), and at $40\times$ more data (-0.04%, p=0.97), across six pose representations and both branch and early-fusion integrations. Meanwhile the no-pose baseline improves 19% from data alone, and the pose *branch* costs a consistent $\sim 1.6\%$ whether or not the poses it receives are the right ones.
- **A directed search does not find one either.** We let a coding agent rewrite the pose-integration code as the variation operator of an evolutionary search over a frozen training repository, with a held-out evaluation split the loop never sees. Two rounds and 20 scored candidates accepted none. The most promising candidate reached -1.21% on the held-out split under a truncated 12-epoch proxy; at the full 30-epoch recipe with n=8 seeds it measured -0.04% (p=0.92)—a convergence-rate artifact, and the tenth self-caught artifact in this project.
- **The null has a mechanism, and it generalizes.** Body facing predicts which way a standing pedestrian steps off at roughly twice chance—but the tracker's box heading estimates the same quantity better (25.2° vs. 44.2° median error), is already an input channel to the model, and is not beaten by pose in any range, occlusion or heading-jitter stratum. Scoring the pose estimator per joint against ground-truth keypoints shows the head carries a tenth of the skill of any limb joint while its MPJPE looks *better* than either wrist—so MPJPE rank-orders joints roughly backwards with respect to information. We close with the condition an auxiliary modality must satisfy to pay at all.

2 Related Work

2.1 Trajectory Forecasting for Autonomous Driving

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 Human Pose Estimation and Pose Representations

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization and is widely used to train human motion priors (Rempe et al., 2021; He et al., 2022). Waymo-3DSkelMo (Zhu et al., 2025) combines LiDAR-based human mesh recovery (Fan et al., 2023) with such priors to produce temporally coherent SMPL pose sequences for Waymo pedestrians; we use its output as pose pseudo-ground-truth. Works such as MotionBERT (Zhu et al., 2023) and HumanMAC (Chen et al., 2023) focus on pose quality but do not connect body pose to downstream trajectory prediction. A separate line uses 2D pose for pedestrian *crossing-intention* classification (Fang & López, 2018; Rasouli et al., 2019), supporting the premise that body configuration carries intent, though without producing metric trajectory forecasts.

2.3 Joint Pose and Trajectory Models

Several recent works condition trajectory prediction on body pose. Salzmann et al. (2023) leverage 3D skeletal keypoints from RGB-D cameras to improve trajectory prediction in robot navigation scenarios. Saadatnejad et al. (2024) integrate 2D/3D skeleton pose cues into a transformer-based predictor on pedestrian benchmarks, and Gao et al. (2025) extend this line to social navigation datasets using skeleton-based body language features. In contrast, our work uses parametric SMPL body pose in 6D rotation space—a richer, anatomically-grounded representation—within the state-of-the-art MTR framework on Waymo, and, with seed-replicated experiments and per-pedestrian confidence intervals, isolates which design choices actually drive the gain: auxiliary pose-loss design, temporal encoding (GRU vs. cross-attention $\pm$ positional encoding), future-pose supervision quality, and the interaction of pose with HD map context and backbone pretraining.

3 Method

3.1 Background: Motion Transformer (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 Pose Conditioning Branch

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \ldots, \mathbf{p}_T \in \mathbb{R}^{144}$, a two-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \text{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \tag{1}$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

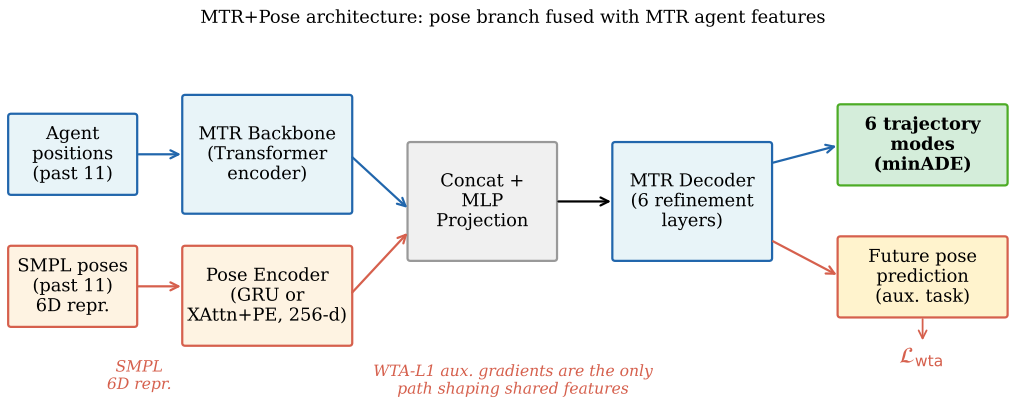


Figure 1: **MTR+Pose architecture.** A GRU pose encoder processes the past 11 SMPL body poses (each represented as a 144D 6D-rotation vector) and produces a 256D hidden state that is concatenated with the MTR agent feature and projected before the decoder. Auxiliary future-pose prediction heads are trained with a winner-takes-all L1 loss $\mathcal{L}_{\text{wta}}$ on 6D rotations; gradients from this auxiliary task are the only path by which pose supervision shapes the shared features. The trajectory decoder and map attention modules of MTR are unchanged.

Pose Prediction Heads. Each of the decoder layers additionally predicts K modes of future poses: at layer ℓ and mode k, a linear head outputs 6D rotations for all 80 future timesteps (a 80×144 output). These heads are auxiliary—their predictions never feed back into the trajectory pathway; their only effect on trajectory prediction is through the gradients their losses send into the shared decoder features and the GRU encoder.

Auxiliary Pose Losses. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i. We consider four auxiliary signals:

- **Winner-takes-all L1 (WTA-L1)**: $\mathcal{L}_{\text{wta}}$ selects, per pedestrian, the pose mode closest to the ground truth in 6D L2 distance, and applies an L1 regression loss to that mode, masked by per-timestep trajectory validity. (In our configuration files this loss is named `gmm_pose` for historical reasons; it is a plain WTA regression, not a Gaussian mixture likelihood.)
- **Geodesic loss**: $\mathcal{L}_{\text{geo}} = \frac{1}{IJ}\sum_{i,j} \arccos\Big(\text{clamp}\Big(\frac{\text{tr}(\mathbf{R}_i^{j\top}\hat{\mathbf{R}}_i^j)-1}{2}, -1{+}\epsilon, 1{-}\epsilon\Big)\Big)$, where $\mathbf{R}, \hat{\mathbf{R}} \in SO(3)$ are rotation matrices recovered from the 6D representations of the WTA-selected mode, averaged over all future steps (unmasked). The clamp ($\epsilon{=}10^{-7}$) avoids the divergent $\arccos$ derivative at the ±1 boundary.
- **MPJPE**: $\mathcal{L}_{\text{mpjpe}} = \frac{1}{IJ}\sum_{i,j} \|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.
- **Pose mode classification**: $\mathcal{L}_{\text{cls_pose}}$, a cross-entropy loss on per-mode pose scores against the WTA-selected mode.

The total loss adds the three MTR motion terms ($\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + 0.5\,\mathcal{L}_{\text{vel}}$) to the weighted pose terms $w_{\text{wta}}\mathcal{L}_{\text{wta}} + w_{\text{geo}}\mathcal{L}_{\text{geo}} + w_{\text{mpjpe}}\mathcal{L}_{\text{mpjpe}} + w_{\text{cls_pose}}\mathcal{L}_{\text{cls_pose}}$; we ablate $w_\star \in \{0, 0.05, 0.1, 0.2\}$ (weights of 1.0 diverge to NaN through the SMPL forward pass, so all stable runs use $w \le 0.2$).

A caveat discovered post-hoc: future pose ground truth is absent in the main data condition. In our primary (10 fps) data condition, SMPL pose observations cover only the *past* window ($t \le 1.0$ s); future pose targets are zero for 99.99% of timesteps (5 of 91,520 in the training split). As we show analytically and empirically in Section 4.8, this renders the geodesic term *inert* (identically zero gradient against all-zero targets, which map to the all-zero matrix under Gram–Schmidt), while WTA-L1, MPJPE, and $\mathcal{L}_{\text{cls_pose}}$ stay active but pull predictions toward the degenerate zero targets. We report it prominently because it reverses a conclusion an earlier draft drew from single-run comparisons.

3.3 Cross-Attention Pose Encoders (Ablation)

To isolate the contribution of temporal ordering from sequential integration, we test two cross-attention pose encoder variants. Let $\mathbf{q} \in \mathbb{R}^{256}$ be the MTR agent feature (projected query) and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{11\times256}$ be the 11 projected pose frames (keys and values). The cross-attention output is:

$$\mathbf{c} = \mathrm{softmax}\left(\frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{256}}\right)\mathbf{V}, \tag{2}$$

and $\mathbf{c}$ is concatenated with $\mathbf{q}$ and projected to 256D, replacing $\mathbf{h}_T$ in the fusion step.

Cross-attention without positional encoding. No temporal information is injected before attention: $\mathbf{K} = \{W_K\mathbf{p}_t\}_{t=1}^{11}$, $\mathbf{V} = \{W_V\mathbf{p}_t\}_{t=1}^{11}$. This treats pose frames as a permutation-invariant set and tests whether the raw content of each pose frame—absent any ordering—is sufficient to condition trajectory prediction.

Cross-attention with sinusoidal positional encoding. Standard sinusoidal positional encoding (Vaswani et al., 2017) is added to the projected keys and values before attention: $\mathbf{K}' = \{W_K\mathbf{p}_t + \mathbf{e}_t\}_{t=1}^{11}$, where $\mathbf{e}_t$ is the fixed sinusoidal embedding for frame index $t \in \{0, \ldots, 10\}$. This injects temporal ordering at zero extra parameter cost and tests whether ordering alone is sufficient, without the causal accumulation structure of a GRU.

4 Experiments

4.1 Dataset and Setup

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body pose pseudo-ground-truth from the Waymo-3DSkelMo pipeline (Zhu et al., 2025). The subset contains 579 training scenes (1,348 pedestrians, of which 1,144 have sufficient valid past observations to serve as prediction targets) and 5,171 validation scenes (8,282 pedestrians). Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10 Hz; 9.0 s total span — 1.0 s past plus an 8.0 s future horizon). 6,688 of the 8,282 validation pedestrians (80.8%) have valid future trajectory data and enter the evaluation.

Pose provenance. The SMPL poses are not motion-capture ground truth: they are estimated from Waymo LiDAR by the Waymo-3DSkelMo pipeline (Zhu et al., 2025), which fits per-frame SMPL bodies with LiDAR-HMR (Fan et al., 2023) and refines them with AMASS-trained motion priors (Mahmood et al., 2019; Rempe et al., 2021; He et al., 2022). Two properties matter for our results. *First*, pose observations exist only for the past window in our main condition, so no future information leaks into the inputs; the evaluation is sound. *Second*, the pipeline snaps the SMPL root rotation to the past-trajectory motion direction whenever the pedestrian is moving and the raw LiDAR estimate deviates by more than 90°, so the root-orientation channel is partially derived from position history the model already receives—an overlap we quantify in Section 5. Body configuration and gait within $\pm90^\circ$ are genuinely perception-derived.

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.0 s future horizon (80 steps at 10 Hz). As an externally interpretable anchor we also compute **MR@2m** (fraction of pedestrians with best-checkpoint minFDE > 2 m): 0.239 no-map, 0.145 map, 0.093 pretrained; pose moves it by ≤ 0.005 in any cell. Official Waymo metrics (mAP, gated miss rate) require full retained predictions, which our per-epoch protocol does not store.

Training and statistical protocol. All models train for 30 epochs with batch size 10 on a single RTX 4090 (AdamW, LR 10^{-4}, step decay $\times0.5$ at epochs 10/20/25, gradient-norm clip 1.0); a full run takes 10–15 minutes. Validation is run at epochs 1, 2, 4, every second epoch up to 20, and every epoch from 21 to 30. Following common practice, our per-run headline number is the *best* validation minADE over these evaluations; because best-of-n selection on a noisy curve inflates apparent differences, every key comparison is additionally run with **three to five random seeds** (five for the headline, encoder, map, and pretrain cells; three for secondary cells), and we report mean $\pm$ std across seeds together with **paired bootstrap** 95% confidence intervals over the 6,688

Table 1: Clean core. minADE (m) on the fixed validation set; n=5 seeds in a fixed order. Relative differences with hier p in parentheses; negative = better for the first-named arm. —: reference arm or not tested.

Arm	Per-seed minADE	Mean ± sd	vs. no-pose	Content test
From scratch				
No-pose	.4584 .4428 .4530 .4954 .4500	0.4599 ± 0.0206	—	—
Shuffled	.4544 .4668 .4514 .4547 .4672	0.4589 ± 0.0075	-0.21% (0.96)	—
Pose	.4705 .4557 .4530 .4570 .4736	0.4620 ± 0.0094	$+0.45\%$ (0.76)	$+0.66\%$ (0.46)
Fine-tuned from a full-Waymo trajectory-only backbone				
No-pose	.3721 .3658 .3670 .3727 .3684	0.3692 ± 0.0031	—	—
Shuffled	.3637 .3627 .3644 .3744 .3726	0.3676 ± 0.0055	-0.44% (0.46)	—
Pose	.3639 .3723 .3708 .3739 .3732	0.3708 ± 0.0040	$+0.44\%$ (0.49)	$+0.88\%$ (0.11)

evaluated pedestrians (per-pedestrian errors averaged across seeds, resampled jointly for both models in a pair; 10,000 resamples).

Scale caveat. The 579-scene training subset is a small fraction of the full Waymo training set (≈487k scenarios), so absolute minADE values in Sections 4.3–4.7 are far from full-data state of the art; relative, seed-replicated comparisons between models trained on identical data are the object of study. Section 4.4 quantifies the scale effect directly.

4.2 Reading the Tables: the Comparison That Matters

Every number below is computed on the corrected per-scene re-export (Section 4.1); nothing from the id-collided export is reused. The validation set is fixed throughout: 5,171 scenes and 16,486 evaluated pedestrians. There are three arms. *No-pose* receives trajectories and map only. *Pose* adds the pedestrian's own SMPL sequence. *Shuffled* is the identical architecture fed *another pedestrian's* SMPL sequence.

Pose versus no-pose confounds the information in the body signal with the capacity of an extra input branch; pose versus shuffled holds parameters, optimiser, input dimensionality and recipe fixed, and isolates content. We call the latter the *content test* and treat it as decisive. Relative differences are $(A - B)/B$ for "A vs. B"; negative means A is better. The primary test is a hierarchical bootstrap resampling seeds and then pedestrians (hier p); a pedestrian-only paired bootstrap ignores between-seed variance, is anti-conservative, and is never used as a gate.

4.3 Clean Core

Table 1 is the corrected version of the configuration behind our own retracted claim: 579 training scenes (6,348 training pedestrians), five seeds per arm, 30 epochs, map context on, cross-attention+PE encoder, in two regimes. *Fine-tuned* initialises every arm from one trajectory-only backbone pre-trained on full Waymo (231,507 examples from 486,995 scenarios) and attaches the pose branch through a residual, zero-initialised fuser, so both pose arms begin as the no-pose model.

Neither regime shows an effect, and in both the shuffled arm has the lowest mean of the three: another pedestrian's body is nominally the best input. Fine-tuning lowers every arm and tightens the seed spread (sd 0.0031–0.0055 against 0.0075–0.0206), and the content test stays null with its sign against pose.

Seed count decides what this table appears to say. With the first three seeds only, the same design read as *pose harms* from scratch ($+1.85\%$, hier p=0.037) and *shuffled is best* after fine-tuning (-1.27% vs. no-pose, hier p=0.016). Both dissolved at n=5. Seed 404's from-scratch no-pose run (0.4954)—the worst single run in the table—lies outside the three-seed set, and its absence flattered the baseline.

Table 2: Scale ladder. Validation is identical at every rung (5,171 scenes / 16,486 pedestrians). minADE mean ± sd over seeds; relative differences with hier p; negative = better for the first-named arm.

Training scenes / pedestrians	579 / 6,348	5k / 31,302	25k / 142,807
Learning rate	10^{-4}	5×10^{-5}	2.5×10^{-5}
Seeds	5	3	5
No-pose	0.4599 ± 0.0206	0.4176 ± 0.0060	0.3716 ± 0.0027
Pose	0.4620 ± 0.0094	0.4138 ± 0.0053	0.3778 ± 0.0024
Shuffled	0.4589 ± 0.0075	0.4181 ± 0.0068	0.3777 ± 0.0053
Pose vs. no-pose	$+0.45\%$ (0.76)	-0.90% (0.52)	$+1.66\%$ (0.0001)
Shuffled vs. no-pose	-0.21% (0.96)	$+0.12\%$ (0.85)	$+1.62\%$ (0.0082)
Content test	$+0.66\%$ (0.46)	-1.02% (0.44)	$+0.04\%$ (0.97)

4.4 Scale Ladder

If pose content is real but 579 scenes are too few to exploit it, more data should open a gap between the pose and shuffled arms. Table 2 trains all three arms from scratch at three rungs, with validation held identical and the learning rate selected *per arm* at each rung.

Data alone moves the no-pose baseline from 0.4599 to 0.4176 to 0.3716 (-19.2%). The content test is null at every rung. At 25k the bootstrap interval on the absolute content difference is $[-0.0040, +0.0047]$ m, about $\pm1.2\%$ relative: at the largest rung we train, the data exclude a content effect beyond roughly 1.2% in either direction.

The branch-capacity tax. At 25k both pose arms are significantly worse than no-pose—pose $+1.66\%$ (p=0.0001), shuffled $+1.62\%$ (p=0.0082)—and indistinguishable from each other ($+0.04\%$, p=0.97). The branch costs about 1.6%, and it costs the same whether it carries the right body or the wrong one. A pose-versus-no-pose comparison at this rung would report a significant *harm* from pose; the content test shows the harm is architecture, not content.

The tuning phantom. The 5k rung first read as catastrophic. With the learning rate inherited from the 579 recipe (10^{-4}, one seed) pose was $+37.1\%$ versus no-pose; at 5×10^{-5} for eight epochs, $+7.4\%$; with per-arm selection and full training, -0.90%, n.s. The 25k per-arm sweep (four rates $\times$ three seeds $\times$ three arms) gives every arm an interior optimum at 2.5×10^{-5} and moves minADE by up to 16.7% across the grid, against a content signal of 0.04–0.09%: the learning rate is roughly $190\times$ the treatment. A pose-versus-baseline claim that does not tune each arm separately is a claim about the learning rate. (The eight-epoch and grid values come from training logs rather than our archived evaluation outputs.)

4.5 Representation and Integration

The null could be a property of how pose enters the model rather than of what it contains. We tested the three ways that could be so.

Coordinate frame. Trajectories are agent-centric, but the SMPL root orientation had remained in the world frame, so body facing *relative to travel* was not expressible from the input. The defect touches 6 of 144 dimensions of the 6D representation and all 72 dimensions of joint positions. After rotating the root into the agent frame, the median |body facing − travel heading| is 12.0° for correct pose against 93.4° for shuffled—the first configuration in this study in which the control is separable from the treatment *at the input*. It is not separable at the output: the frame-fixed branch (n=3) scores 0.3794 ± 0.0024 against its own shuffled control 0.3768 ± 0.0071, a content test of $+0.68\%$ (hier p=0.37), with $+2.23\%$ versus seed-matched no-pose (p=0.0006)—the branch tax again.

Early fusion, no branch. To remove the branch tax rather than pay it, we replaced the 144-d branch with eight frame-invariant descriptors (facing-vs-motion, torso yaw rate, stride phase, cadence, arm

Table 3: Pre-registered confirmation: three arms × eight seeds × 30 epochs; paired t over seeds; negative = better for the first-named arm. The primary endpoint is the first row. †: the most-selected cell *and* a capacity-confounded cross-architecture comparison whose own content control is n.s.; not a win.

Split	Epoch	Content test	Pose vs. no-pose
Final (held out)	last (primary)	-0.04%, $t(7)=-0.11$, $p=0.916$, $5/8$	$+0.88\%$, $p=0.179$
Final (held out)	chosen on dev	-0.19%, $p=0.63$	-0.46%, $p=0.36$
Dev	last	$+0.29\%$, $p=0.25$	$+0.97\%$, $p=0.08$
Dev	best on dev	-0.29%, $p=0.34$	-0.44%, $p=0.030^{\dagger}$

swing, spine tilt) appended to the agent input channels ($29 \to 37$), so pose enters through the existing encoder. Result: 0.3711 ± 0.0051; -0.01% versus seed-matched no-pose (hier $p=0.9988$); -1.66% versus the 144-d branch design ($p=0.0056$); content test -0.48% ($p=0.65$). Pose can be carried for free. It still does not pay.

A directed search over integrations. Hand-designed integrations sample a small space. We therefore ran a directed search in which a coding agent rewrites the pose-integration code as the variation operator of an evolutionary loop against a frozen, commit-pinned training repository, scoring on a dev split and never seeing a held-out final split. A candidate is accepted only if it improves aggregate minADE by at least 0.8%. Round 1 scored 13 candidates and accepted none, its best improving minADE by 0.49%. Round 2 widened the input to 216 dimensions by adding canonical-frame joint positions, scored seven and accepted none, its best improving by 0.76%, below the floor. Two module-mode candidates reproduced the branch tax, worsening minADE by 1.44% and 4.66%.

4.6 The Pre-registered Confirmation

Round 2's best candidate is the strongest pose-content signal this study produced, so we pre-registered its confirmation. Re-scored at five seeds against its own shuffled control under the 12-epoch scoring proxy, it read -1.07% on dev ($t(4)=-3.01$) and -1.21% on the held-out final split ($t(4)=-4.24$, pose better in $5/5$ seeds). Before running anything further we fixed the protocol: three arms × eight seeds × 30 epochs; primary endpoint the final split at the fixed last epoch; content test; paired t over seeds; $\alpha=0.05$.

The primary endpoint returns -0.04%, $p=0.916$, pose better in 5 of 8 seeds: **not confirmed**. The 12-epoch signal was a convergence-rate artifact—the arms differ in how fast they train, not in where they end.

One cell is nominally significant: dev split, best epoch chosen on dev, pose versus no-pose, -0.44%, $p=0.030$. It is the most-selected cell in the table (the epoch is chosen on the split it is scored on), it is the capacity-confounded cross-architecture comparison, and its own content control in the same cell is -0.29%, $p=0.34$. We report it so that it cannot be quoted as a win.

4.7 Where a Benefit Could Still Hide

An aggregate null can hide a benefit local in time, in behaviour, or in a subpopulation. We looked in the three places a body-pose prior should show first, on the eight confirmation seeds and the held-out split.

Horizon. The content test is $+0.48\%$ at 2 s ($p=0.32$), $+0.24\%$ at 4 s ($p=0.52$), -0.04% at 8 s ($p=0.92$) and -0.04% on minFDE ($p=0.95$): null at every horizon and *least* favourable at the shortest. At 2 s minADE is 0.091 m over roughly 2.8 m of travel.

Manoeuvre. For the $6{,}764$ validation agents with motion covariates we ran the content test within each quartile and the top decile of four covariates: residual from constant-velocity extrapolation, heading variation, speed range, and peak acceleration. Of 20 tests, zero are significant. The hardest decile by constant-velocity residual (mean minADE 1.04 m) gives -0.13%, $p=0.80$.

Table 4: Pose supervision ablation, seed-replicated (no map, GRU encoder, w=0.1 for active losses). "No aux" has no active auxiliary pose loss (the inert geodesic term only), so the pose encoder is trained by trajectory gradients alone. The WTA-L1 row pools the draft's "geodesic-only" and "WTA-L1 only" cells, which are functionally identical. Single-run draft values shown for comparison.

Aux. pose loss	minADE (mean±std)	per-seed values	single-run (draft)
None (no pose)	0.6604 ± 0.0143	.655 .683 .664 .654 .646	0.6745
No aux (inert geo)	0.6545 ± 0.0271	.696 .665 .639 .626 .647	—
WTA-L1 (10 seeds, pooled)	0.6957 ± 0.0841	.639 .785 .629 .717 .657 .897 .662 .662 .659 .650	0.6467 / 0.6231[†]
MPJPE + WTA-L1	0.6723 ± 0.0321	.724 .662 .644 .680 .651	0.6576
All (MPJPE+WTA+cls)	0.6618 ± 0.0366	.668 .674 .630 .623 .714	0.6532

[†]The earlier draft reported these as two different configurations ("WTA-L1 only" and "geodesic-only"); they are the same configuration, and their spread is seed noise.

Subpopulation. Standing-then-moving pedestrians—stationary in the observed history, moving in the future—are the canonical case for a body-pose prior: the trajectory carries no velocity and the body is supposed to carry the intent. On the full validation set they number 1,809. Content test: $+0.38\%$, p=0.80.

Across Sections 4.3–4.7 the content test is null in every configuration, at every training scale, at every horizon, in every manoeuvre stratum, and in the subpopulation built to favour it. The only effects that survive the seed count are the branch-capacity tax, in three forms, and one selection artifact.

4.8 The Auxiliary-Loss Audit: an Inert Term Credited with a Headline Gain

An earlier draft ablated the pose supervision with one run per configuration and concluded geodesic-only supervision was the critical ingredient (0.6231, -7.6% vs. baseline 0.6745), beating "WTA-L1 only" (0.6467). Three facts reverse that conclusion.

Fact 1: the geodesic term received zero gradient. In the 10 fps data condition, future pose targets are zero for all but 5 of 91,520 training steps (Appendix A.2). A zero 6D target maps under Gram–Schmidt to the all-zero matrix $\mathbf{R}_{\text{gt}}$=$\mathbf{0}$, for which $\text{tr}(\hat{\mathbf{R}}^\top \mathbf{R}_{\text{gt}}) \equiv 0$ for *any* prediction, so $\mathcal{L}_{\text{geo}} \equiv \arccos(-\frac{1}{2})$, a constant; we verified its gradient is exactly zero (and nonzero under random targets).

Fact 2: the two compared configurations were therefore identical. The "geodesic-only" run also carried WTA-L1 at w=0.1, so with the geodesic term inert it ran the same training distribution as "WTA-L1 only"—their 0.0236 gap (0.6231 vs. 0.6467) is pure run-to-run noise, larger than most effects the draft reported.

Fact 3: on the corrected data the losses are inert *by construction*. The re-export stores past-only sequences (at most 11 frames), so the future-pose validity mask is empty and all four auxiliary pose losses evaluate to exactly 0.0; we confirmed this by instrumenting each term during training. The consequence is stronger than "ineffective": on this data an auxiliary-loss cell differs from the no-pose baseline only by possessing a pose *encoder*. Any auxiliary-loss comparison we or others report on WOMD-derived pose of this form is a comparison of encoders, not of supervision. Table 4 records the wrong-pose-era measurements that first exposed the noise floor; we retain it as documentation of the artifact, not as evidence about pose.

The corrected interpretation reverses the draft on a second point as well: far from being "the critical ingredient", the auxiliary losses are—with the GRU encoder and degenerate zero-valued targets—associated with *instability*. Four of fifteen runs across the WTA-L1 and MPJPE+WTA-L1 cells converged badly (≥ 0.71), whereas the no-auxiliary control produced no such seed in five runs. At our statistical power the active-loss variants cannot be usefully ranked against one another, and the earlier draft's mechanistic story about geodesic gradients shaping the GRU was fitted to noise.

Table 5: Per-joint recoverability of LiDAR-based HMR against ground-truth keypoints (1,789 frames over 534 pedestrians, 113 Perception validation segments). *MP* is the error of always predicting the mean root-aligned pose, which equals the joint's ground-truth spread by construction; SKILL $= 1 - \text{MPJPE}/\text{MP}$, with a 95% bootstrap CI over frames. *ratio* is predicted spread over ground-truth spread. The head carries an order of magnitude less skill than any limb joint—yet its MPJPE is *better* than either wrist or ankle.

Joint	MPJPE (mm)	MP (mm)	SKILL	95% CI	ratio
Left elbow	91.2	252.9	+0.639	[+.623, +.655]	0.98
Right elbow	91.5	248.9	+0.632	[+.616, +.647]	0.99
Left hip	39.8	97.9	+0.594	[+.582, +.604]	0.70
Right hip	39.8	97.9	+0.593	[+.582, +.604]	0.70
Left wrist	118.3	271.1	+0.564	[+.545, +.581]	0.96
Right wrist	125.4	276.4	+0.546	[+.527, +.565]	0.99
Left shoulder	87.3	171.5	+0.491	[+.474, +.507]	1.07
Right ankle	114.0	221.6	+0.485	[+.464, +.505]	0.86
Right knee	78.4	150.8	+0.480	[+.463, +.496]	0.91
Left ankle	116.6	223.8	+0.479	[+.458, +.498]	0.84
Left knee	81.5	151.2	+0.461	[+.444, +.478]	0.87
Right shoulder	89.0	164.0	+0.457	[+.441, +.473]	1.10
Head	**111.6**	**116.9**	**+0.045**	**[+.022, +.067]**	**0.49**
Overall	91.1	188.1	+0.516		

The mask fix tames the worst failures, but adds no signal. Because the audit shows the pose losses train against absent targets, we re-ran the audit cells (five seeds each) with a corrected future-pose validity mask (losses averaged only over the $< 0.01\%$ of steps that carry a real target; this is the released code). Masking removes the WTA-specific catastrophic tail—the masked WTA-L1 cell's worst seed is 0.679 vs. 0.897 unmasked (σ 0.065→0.018)—but does not turn any auxiliary loss into a benefit: at five seeds *no* masked cell differs from the masked baseline under seed resampling (all hierarchical $p \geq 0.15$), degraded seeds (≥ 0.70) still occur even with no active auxiliary, and the fix leaves the no-pose baseline unchanged (0.6605 ± 0.0153 masked vs. 0.6604; p=0.90). The on-10fps auxiliary pose signal is, as predicted, inert; the no-map regime is noisy with or without it.

Reproducibility note. All seed-matrix results were produced with the *pre-fix* code (tagged in the release); the released code masks the future-pose losses, so re-running any aux-active cell yields the masked training distribution. The baseline is verified unchanged (paired p=0.90, hierarchical p=0.99) and the four audit cells are re-run masked above; other aux-active cells were not, and may differ under the fix.

4.9 Pose Quality: Which Joints LiDAR-based HMR Actually Recovers

A null result invites the objection that the input pose was simply too poor to help. Aggregate MPJPE answers that objection badly, because it cannot separate two very different situations: an estimator that tracks a joint imprecisely, and an estimator that ignores the input and emits its *prior*—roughly the mean human pose. The second is what breaks downstream consumers, and it is invisible in MPJPE, since a joint that barely moves scores well merely by predicting the population mean.

We therefore score each joint against the strongest trivial competitor: always predicting the mean root-aligned ground-truth pose. With MPJPE_j the error at joint j and MP_j the error of that constant predictor,

$$\text{SKILL}_j \;=\; 1 - \text{MPJPE}_j/\text{MP}_j,$$

so $\text{SKILL}_j > 0$ means the estimator beats the mean pose and $\text{SKILL}_j \leq 0$ means it contributes nothing at that joint. We evaluate the LiDAR-HMR stage against ground-truth LiDAR keypoints on the Waymo Perception validation split, matching frames by timestamp, over the 13 joints common to both skeletons (Table 5).

Every limb joint clears the mean-pose baseline decisively, cutting its error by 46–64%. The head does not: it cuts error by 4.5%, an order of magnitude less than the weakest limb joint, and its spread

ratio of 0.49—against 0.70–1.10 for the limbs—shows the estimator additionally under-dispersing it. The head's skill is nonetheless significantly positive ($[+.022, +.067]$): the model is not entirely ignoring it, and we make the quantitative claim rather than a null one. (At a fifth of this sample the head's CI did span zero; that CI was a power artifact, and we note it because it is the kind of small-n null this paper elsewhere cautions against.)

The consequence for practice is that **MPJPE rank-orders these joints almost backwards with respect to information content**. By MPJPE the head (111.6 mm) looks better than either ankle (114.0/116.6 mm) and better than either wrist (118.3/125.4 mm), while carrying roughly a tenth of their skill. The head's flattering error reflects how little the estimator improves on the mean there: its MPJPE (111.6 mm) sits just below its mean-pose baseline (116.9 mm), whereas the hips—which have an even smaller ground-truth spread (97.9 mm)—are predicted at 39.8 mm against a 97.9 mm baseline. Small spread alone is therefore not the explanation; what distinguishes the head is that the estimator does not beat the mean despite it. The physical reading is straightforward—a pedestrian at 20–40 m returns a few dozen LiDAR points, and the head is a near-spherical handful of them with little geometry to constrain its orientation, so the fit falls back largely on the prior.

This bounds what pose conditioning could have delivered, and it does so constructively: gait-bearing joints are genuinely measured, so the null in Section 4.3 is not an artifact of unusable input; but any downstream method depending on head or gaze orientation from LiDAR HMR is reading something close to a prior, whatever the reported MPJPE. We recommend reporting per-joint skill alongside MPJPE for this reason.

5 WHY POSE DOES NOT PAY

The null is not a failure of the estimator or of the architecture search. It follows from what pose measures and what forecasting needs.

Pose carries exactly one predictive cue. If body configuration predicts anything a trajectory does not, it is which way a standing pedestrian will step off. We isolated this as a task. From the corrected export we took every pedestrian stationary at the prediction time (speed < 0.3 m/s): $224{,}946$ in training and $19{,}939$ in validation, 99.4% with a fitted pose.[1] Onset—a displacement above 1 m—follows within 3 s for 8.5% of them. On the $1{,}705$ positive events at the 3 s horizon, SMPL body facing predicts onset direction at roughly twice chance: median angular error $44.2°$ and 50.6% within $45°$, against $90°$ and 25% for a uniform guess.

The tracker already supplies that cue, more accurately. Waymo's per-object box heading estimates the same quantity with median error $25.2°$ and 63.9% within $45°$. Paired on the same events, SMPL is $7.3°$ worse in median (Wilcoxon $p{=}2.6 \times 10^{-43}$) and is the better estimate on only 35% of events—even though the SMPL forward axis was calibrated *against* the box heading before scoring, a choice that favours pose. That heading enters the agent encoder as the sine and cosine of the heading channel. Every pose experiment in this paper therefore competed against a channel the model already had, in a more accurate form than mesh fitting recovers it.

Pose does not take over where the tracker degrades. Stratifying the same paired comparison by range to ego, by history validity (an occlusion proxy) and by box-heading jitter, the pose-minus-box gap is positive in all twelve strata (every $p < 10^{-5}$) and *widens* as perception degrades: $+12.3°$ at 2–17 m rising to $+32.9°$ at 41–75 m, and $+16.9°$ on complete tracks rising to $+33.7°$ on gappy ones. In the jitteriest heading quartile, where the box heading is genuinely poor ($50.5°$), SMPL is worse still ($63.8°$). Both estimates derive from the same LiDAR returns; they fail together, not complementarily.

And nothing on top of it. A model predicting onset direction from box heading alone reaches a median of $24.8°$ ($18{,}950$ positive training events, $1{,}688$ test). Appending a full pose descriptor gives $26.5°$ (Wilcoxon $p{=}0.085$ vs. box-only); appending a *shuffled* descriptor gives $25.8°$ ($p{=}0.214$). The small degradation is the cost of extra feature dimensions, not a property of pose content. The

[1] Deduplicated by (scene, object): Waymo writes several sample files per scene carrying identical tracks, so the raw extraction is inflated $5.4\times$.

companion question—whether the pedestrian moves at all—is carried by kinematics: residual micro-motion alone gives test AUC 0.741 / 0.703 / 0.638 at 2 / 3 / 5 s, and pose adds nothing the shuffled control does not add equally.

The one cue that could have been complementary is not measured. Gaze precedes locomotion in human motor control, so head yaw relative to the torso is a plausible cue independent of the box heading. It is not available from this sensor. In the fitted meshes, head-versus-torso yaw has a standard deviation of $3.8°$ (5th–95th percentile $[-4.4°, +5.8°]$), and the head axis-angle varies about six times less than the knee: the estimator is emitting its pose prior. The real quantity is far wider—against ground-truth 3D keypoints on Waymo Perception (n=8,234) its circular standard deviation is $19.2°$, with 25% of pedestrians beyond $15°$. This is a measurement failure, not evidence that gaze is behaviourally irrelevant: a pedestrian at 20–40 m returns a few dozen LiDAR points, and the head is a near-spherical handful of them with little geometry to constrain yaw. The ceiling is low even for an oracle—the residual head yaw would have to explain has a spread of $66.9°$, so a perfect measurement at $19.2°$ covers under a third of it, and the fitted value explains 0.11% of it ($r = -0.033$, p=0.17, wrong sign).

None of this is visible in an aggregate pose error, which is why Section 4.9 scores recoverability per joint: a single MPJPE cannot distinguish an estimator that tracks a joint imprecisely from one that ignores it and emits its prior, since both produce a moderate mean distance.

The general condition. Two results locate where pose could have helped. Across horizons the content test is null everywhere and least favourable at the *shortest*: $+0.48\%$ at 2 s against -0.04% at 8 s. At 2 s minADE is 0.091 m while a pedestrian at $1.4\,\mathrm{m\,s^{-1}}$ travels $\approx$2.8 m, so short-horizon forecasting is very nearly *saturated* by constant-velocity extrapolation and has almost no error left for a second modality to remove; at long horizons the residual is dominated by destination and intent, which body configuration does not disclose. Pose is squeezed out at both ends. Stratifying instead by how badly constant-velocity extrapolation fails—the sharpest operational definition of "not saturated"—gives 20 content tests and zero significant results, including the hardest residual decile (-0.13%, p=0.80).

That second result is the decisive one: it removes the saturation objection and the null survives. We therefore state the principle in the general, falsifiable form that is the most portable thing this paper can offer. *An auxiliary modality improves a forecast only where the baseline is neither saturated nor dominated by intent the modality cannot observe: there must be error left to remove, and its cause must be legible in the new input.* The two clauses are independent, and the manoeuvre strata relax only the first—when a pedestrian departs from constant velocity the cause is typically external (another agent, an obstacle, a kerb, a signal, a destination) rather than written in their own body. The same argument predicts failure for any appearance-derived signal under these conditions, and says where to look instead: regimes where prediction is hard *for reasons visible in the agent itself*.

6 Methodology: ten self-caught artifacts

Every quantitative claim this project made about pose, over eighteen months, was eventually retracted by one of its own checks. We list the mechanisms rather than the incidents, because the mechanisms transfer.

How an inert loss became a headline result. The failure chain is worth spelling out because none of its links is exotic: a data condition silently degenerated (all-zero future-pose targets), the loss evaluated to a smooth constant with *zero* gradient rather than crashing, two functionally identical configurations were each run once, and the resulting 0.024 seed gap was read as a 3.8% design effect and given a mechanistic story. The audit that caught it generalises to two cheap checks: verify the *target coverage* of every auxiliary loss against the data, and check the *gradient* of every loss term in isolation on a real batch.

What caught the rest. *Seed replication* dissolved four apparent effects that were tight at $n \leq 3$: a -7.6% geodesic gain between two identical configurations, a -3.5% "stable encoder" win whose next two seeds both failed, a positional-encoding ordering effect, and—after the data fix—both a "pose harms" and a "shuffled is best" reading of the same triad. *Wrong-input controls* were decisive

twice: a shuffled-pose arm reproduced the one surviving pose benefit, and a wrong-pose control turned a significant intent-classification gain into a null. *Data forensics* produced the largest correction of all—the export keyed pose clips by a scene-local object id, so every pose-conditioned experiment we had run to that point used another pedestrian's body motion. *Protocol discipline* caught three more: a last-epoch checkpoint convention that manufactured a $+13.4\%$ effect (p=0.0002) which became $+4.1\%$ (p=0.37) at best epoch; an inherited learning rate that made pose look catastrophic at a new scale ($+37\%$, falling to -0.9% n.s. once each arm was tuned); and a frame-indexing bug in our own evaluation harness that inflated MPJPE from 87.6 to 160.6 mm. *Pre-registration with a held-out split* caught the last: a candidate that reached -1.21% on data the search never saw, under a truncated proxy recipe, measured -0.04% (p=0.92) at the full recipe—a convergence-rate artifact.

Two statistical lessons. First, a percentile bootstrap over per-seed deltas is badly anti-conservative when the deltas share a sign: at n=5 it returns $p \approx 1/N_{\text{boot}}$ regardless of effect size, and it had manufactured several of the retractions above. We gate on a paired t-test and keep the bootstrap interval as a descriptive spread. Second, best-checkpoint selection is not neutral between architectures: it is worth -1.87% to a wide-input arm and -0.55% to the baseline, enough to flip the sign of a cross-architecture comparison. Only the content test—structurally identical arms differing in whether the input is the right pedestrian's—is robust to it, which is the deeper reason it is the comparison we rely on.

At sub-1% effect sizes, everything is a treatment: the learning rate moves minADE by up to 16.7%, the checkpoint rule by $\sim 1.9\%$, and the seed by more than either, against a content signal of 0.04%.

7 Limitations

Scale. The largest rung we train is 25,000 scenes (142,807 pedestrians). The corrected export covers 299,220 training scenes, so the full rung is a compute limitation rather than an annotation one; we state this because an earlier draft of this paper wrongly claimed pose-conditioning at scale was infeasible with current annotations. At n=8 seeds our minimum detectable effect on the content test is $\approx 1.1\%$, so effects below that remain outside our resolution.

Pose provenance. The SMPL pseudo-ground-truth is a pipeline output, not ground truth. We characterise it rather than assume it: root-aligned MPJPE is 87.6 mm at the mesh-recovery stage and 92.7 mm after the motion-prior stages that the predictor actually consumes, with the degradation concentrated in the gait joints (knees $+19$ mm, ankles $+17$/$+25$ mm); an RGB alternative is worse (106.2 mm). Section 4.9 scores recoverability per joint. Deployed online pose estimation would carry different noise.

Single dataset and modality. All results are Waymo, and the poses are LiDAR-derived; WOMD ships no raw imagery, so a camera-based pose or gaze estimate cannot be tested at this scale. We also do not model pose–pose interaction: each agent's pose is summarised independently, so social cues that live between pedestrians rather than within one are outside what we tested.

Statistics. Our primary test is a hierarchical bootstrap resampling seeds then pedestrians; a pedestrian-only paired bootstrap on seed-averaged errors is anti-conservative and we do not use it as a gate. For the pre-registered confirmation the gate is a paired t-test over seeds, because a percentile bootstrap over n=8 same-signed deltas returns $p \approx 1/N_{\text{boot}}$ irrespective of effect size—a failure mode that had previously manufactured results in this project. We test a large family of pairs without correction; the trajectory-only effects (map, pretraining) survive any reasonable correction, and no pose effect is close. Best-checkpoint selection systematically favours wider-input arms (Section 4.6), which is why cross-architecture comparisons are reported at a fixed epoch and the content test—structurally identical arms—is the one we rely on.

8 Conclusion

We have presented MTR+Pose, a pose-conditioned extension of the Motion Transformer over past SMPL body poses, and used it to ask what body pose is worth for pedestrian trajectory forecasting. The answer is nothing measurable, and the value of the paper is in how tightly that answer is bounded and why it holds.

Our conclusions. (1) **The null is robust and controlled.** On a corrected pose export, correct pose is statistically indistinguishable from the same architecture fed another pedestrian's pose—from scratch (-0.66%, p=0.46), after full-Waymo pretraining (-0.88%, p=0.11), and at $40\times$ the data (-0.04%, p=0.97)—across six representations, branch and early-fusion integrations, a directed agentic search over the integration code, and a pre-registered n=8 confirmation on a held-out split (-0.04%, p=0.92). The pose *branch* costs $\sim$1.6% whether or not its input is the right pedestrian's; early fusion removes that cost but adds no benefit, so pose can be carried for free and still does not pay.

(2) **The null has a mechanism.** The one predictive cue in body pose—facing, for pedestrians about to step off—is already supplied to the model by the tracker's heading channel, more accurately than mesh fitting recovers it (25.2° vs. 44.2°), in every range, occlusion and jitter stratum, and with nothing added on top. Head orientation, the one cue that could have been complementary, is not recovered from LiDAR at all: fitted head-versus-torso yaw varies 3.8° against 19.2° in ground truth.

(3) **Aggregate pose-quality metrics hide this.** Scored per joint against a mean-pose baseline, the head carries a tenth of the skill of any limb joint while its MPJPE looks better than either wrist; MPJPE rank-orders joints roughly backwards with respect to information content. We recommend reporting per-joint skill alongside it.

(4) **An auxiliary modality can only pay where the baseline is neither saturated nor dominated by unobservable intent.** For pedestrian forecasting on Waymo that window is empty, at both ends and in every manoeuvre stratum we tested. The prediction extends to any appearance-derived signal evaluated under the same conditions.

The methodology, not the pose numbers, is our intended lasting contribution: ten self-caught artifacts, three of them our own successive headline claims, caught by seed replication, wrong-input controls, pre-registration, held-out splits, fixed-epoch endpoints and data forensics. At sub-1% effect sizes, everything is a treatment.

Reproducibility Statement

All experiments use the publicly available Waymo Open Motion Dataset and the open-source Waymo-3DSkelMo pose-extraction pipeline; neither is redistributed by us. Training configurations (YAML), the multi-seed launch script, the per-pedestrian metric logger, and the bootstrap analysis script are included in our code release, together with the exact random seeds (101/202/303, plus 404/505 for the headline, encoder, map, and pretrain cells) and the per-epoch validation protocol described in Section 4.1. Appendix A lists all hyperparameters; Appendix A.2 documents dataset construction, including the timestamp-alignment and future-pose-coverage details on which the inert-gradient analysis rests. We will release the code upon publication.

Acknowledgments

The authors gratefully acknowledge access to GPU compute (RTX 4090) and the publicly available Waymo Open Motion Dataset.

LLM Disclosure

The authors used large language model (LLM) assistance (Claude, Anthropic) during the preparation of this manuscript for writing drafts, code assistance, and literature search support. All experimental results, scientific analysis, and conclusions are the authors' own.

References

Ling-Hao Chen, Jiawei Zhang, Yewen Li, Yiren Pang, Xiaobo Xia, and Tongliang Liu. HumanMAC: Masked motion completion for human motion prediction. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9510–9521. IEEE, October 2023. doi: 10.1109/iccv51070.2023.00875. URL http://dx.doi.org/10.1109/iccv51070.2023.00875.

Scott Ettinger, Shuyang Cheng, Benjamin Caine, Chenxi Liu, Hang Zhao, Sabeek Pradhan, Yuning Chai, Ben Sapp, Charles Qi, Yin Zhou, Zoey Yang, Aurelien Chouard, Pei Sun, Jiquan Ngiam,

Vijay Vasudevan, Alexander McCauley, Jonathon Shlens, and Dragomir Anguelov. Large scale interactive motion forecasting for autonomous driving : The waymo open motion dataset. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9690–9699. IEEE, October 2021. doi: 10.1109/iccv48922.2021.00957. URL http://dx.doi.org/10.1109/iccv48922.2021.00957.

Bohao Fan, Wenzhao Zheng, Jianjiang Feng, and Jie Zhou. LiDAR-HMR: 3D human mesh recovery from LiDAR. *arXiv preprint arXiv:2311.11971*, 2023.

Zhijie Fang and Antonio M. López. Is the pedestrian going to cross? answering by 2d pose estimation. In *2018 IEEE Intelligent Vehicles Symposium (IV)*, pp. 1271–1276, 2018. doi: 10.1109/IVS.2018.8500413.

Yang Gao, Saeed Saadatnejad, and Alexandre Alahi. Social-pose: Leveraging body language for pedestrian trajectory prediction. *IEEE Transactions on Intelligent Transportation Systems*, 2025. URL https://arxiv.org/abs/2507.22742. arXiv:2507.22742.

Agrim Gupta, Justin Johnson, Li Fei-Fei, Silvio Savarese, and Alexandre Alahi. Social GAN: Socially acceptable trajectories with generative adversarial networks. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2255–2264. IEEE, June 2018. doi: 10.1109/cvpr.2018.00240. URL http://dx.doi.org/10.1109/CVPR.2018.00240.

Chengan He, Jun Saito, James Zachary, Holly Rushmeier, and Yi Zhou. NeMF: Neural motion fields for kinematic animation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: a skinned multi-person linear model. *ACM Transactions on Graphics*, 34(6):1–16, October 2015. ISSN 1557-7368. doi: 10.1145/2816795.2818013. URL http://dx.doi.org/10.1145/2816795.2818013.

Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael Black. AMASS: Archive of motion capture as surface shapes. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 5441–5450. IEEE, October 2019. doi: 10.1109/iccv.2019.00554. URL http://dx.doi.org/10.1109/iccv.2019.00554.

Amir Rasouli, Iuliia Kotseruba, Toni Kunic, and John K. Tsotsos. PIE: A large-scale dataset and models for pedestrian intention estimation and trajectory prediction. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6261–6270, 2019. doi: 10.1109/ICCV.2019.00636.

Davis Rempe, Tolga Birdal, Aaron Hertzmann, Jimei Yang, Srinath Sridhar, and Leonidas J. Guibas. HuMoR: 3D human motion model for robust pose estimation. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11468–11479, 2021. doi: 10.1109/ICCV48922.2021.01129.

Saeed Saadatnejad, Yang Gao, Kaouther Messaoud, and Alexandre Alahi. Social-transmotion: Promptable human trajectory prediction. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2312.11455.

Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. *Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data*, pp. 683–700. Lecture Notes in Computer Science. Springer International Publishing, 2020. ISBN 9783030585235. doi: 10.1007/978-3-030-58523-5_40. URL http://dx.doi.org/10.1007/978-3-030-58523-5_40.

Tim Salzmann, Lewis Chiang, Markus Ryll, Dorsa Sadigh, Carolina Parada, and Alex Bewley. Robots that can see: Leveraging human pose for trajectory prediction. *IEEE Robotics and Automation Letters*, 8(11):7090–7097, November 2023. doi: 10.1109/LRA.2023.3312035. URL https://doi.org/10.1109/LRA.2023.3312035.

Shaoshuai Shi, Li Jiang, Dengxin Dai, and Bernt Schiele. Motion transformer with global intention localization and local movement refinement. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (eds.), *Advances in Neural*

Information Processing Systems, volume 35, pp. 6531–6543. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/2ab47c960bfee4f86dfc362f26ad066a-Paper-Conference.pdf.

Shaoshuai Shi, Li Jiang, Dengxin Dai, and Bernt Schiele. MTR++: Multi-agent motion prediction with symmetric scene modeling and guided intention querying. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3955–3971, May 2024. ISSN 1939-3539. doi: 10.1109/tpami.2024.3352811. URL http://dx.doi.org/10.1109/tpami.2024.3352811.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Ye Yuan, Xinshuo Weng, Yanglan Ou, and Kris Kitani. Agentformer: Agent-aware transformers for socio-temporal multi-agent forecasting. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9793–9803. IEEE, October 2021. doi: 10.1109/iccv48922.2021.00967. URL http://dx.doi.org/10.1109/iccv48922.2021.00967.

Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5738–5746. IEEE, June 2019. doi: 10.1109/cvpr.2019.00589. URL http://dx.doi.org/10.1109/CVPR.2019.00589.

Guangxun Zhu, Shiyu Fan, Hang Dai, and Edmond S. L. Ho. Waymo-3DSkelMo: A multi-agent 3D skeletal motion dataset for pedestrian interaction modeling in autonomous driving. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 13184–13190, 2025.

Wentao Zhu, Xiaoxuan Ma, Zhaoyang Liu, Libin Liu, Wayne Wu, and Yizhou Wang. MotionBERT: A unified perspective on learning human motion representations. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 15039–15053. IEEE, October 2023. doi: 10.1109/iccv51070.2023.01385. URL http://dx.doi.org/10.1109/ICCV51070.2023.01385.

A Additional Experimental Details

A.1 Architecture Hyperparameters

- GRU pose encoder: GRU(input=144, hidden=256, layers=2, dropout=0.1); ≈505K parameters total including fusion MLP
- Cross-attention pose encoder: pose projection 144→256, MultiheadAttention(embed_dim=256, num_heads=8, dropout=0.1, batch_first=True); ≈497K parameters total including fusion MLP
- Pose input dimension: 144 (24 joints × 6D rotation)
- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR context encoder: 6 local-attention transformer layers, 256 hidden dim, 8 heads, attention dropout 0.1
- MTR decoder: 6 layers, hidden dim 512 (map branch 256), 8 heads
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01
- LR schedule: initial LR 1×10^{-4}, step decay $\times0.5$ at epochs 10, 20, 25
- Epochs: 30

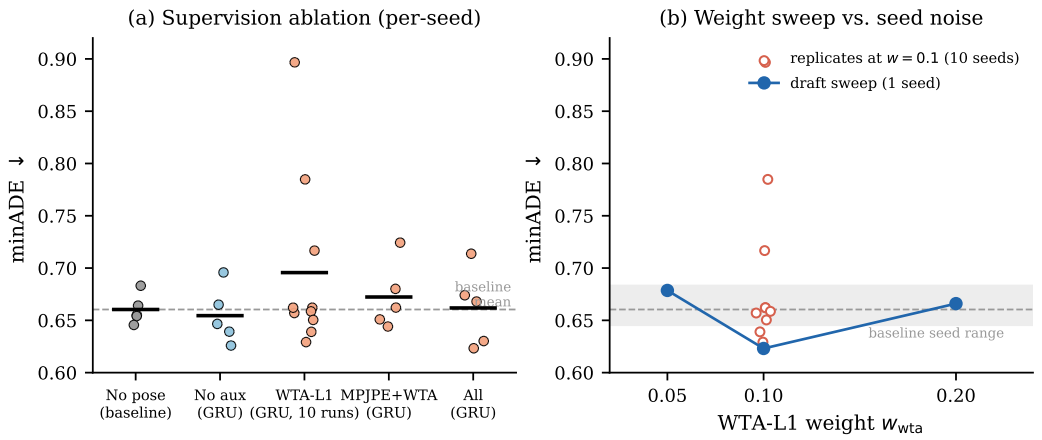


Figure 2: **Loss ablation and WTA-L1 weight sensitivity.** *Left*: minADE for the seed-replicated supervision configurations (Table 4); markers show individual seeds. *Right*: single-seed sensitivity to the auxiliary WTA-L1 weight ($w \in \{0.05, 0.1, 0.2\}$), with the seed-noise scale measured in Section 4.8 shown as a shaded band. Within that noise, the apparent optimum at w=0.1 is weak evidence at best.

- Gradient clipping: norm ≤ 1.0
- Random seeds for replicated runs: 101, 202, 303 (plus 404, 505 for headline cells)

A.2 Dataset Construction

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. 6,688 of 8,282 validation pedestrians (80.8%) have a valid future-trajectory match and are evaluated; the remainder are excluded. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment. In the 10 fps condition the SMPL observations cover only $[0, 1.0]$ s (the past window): across the entire training split only 5 of 91,520 future pose-target steps are nonzero, which is the basis for the inert-geodesic-gradient analysis in Section 4.8.

A.3 Loss Weight Stability

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ± 1 boundary of the $\arccos$ argument; we clamp to $[-1 + \epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 Auxiliary Loss Weight

The earlier draft swept what it believed was the geodesic weight and found a "sharp optimum" at w=0.1: w=0.05 gave 0.6786 and w=0.2 gave 0.6660, both single runs. Two corrections apply. First, because the geodesic term is inert, the sweep actually varied the *WTA-L1* weight (both weights were scaled together in those configurations). Second, the differences involved (0.02–0.06) overlap the seed-noise scale measured in Section 4.8, so the curvature of this one-seed sweep should not be read as a calibrated sensitivity estimate. What survives: w=0.1 is a reasonable default, and we did not find evidence that performance is improved by halving or doubling it.

A.5 Ablation: Future Pose Supervision Quality

To localize the source of improvement, we compare two data conditions, both with the full auxiliary loss set at w=0.1, **seed-replicated (five seeds each)**: (a) *no future poses* — future-pose GT is zero,

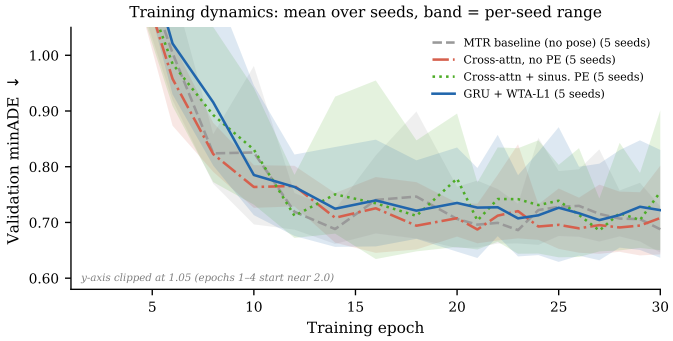


Figure 3: **Training dynamics.** Validation minADE over training for the temporal encoding variants; lines are means over seeds, bands the per-seed range. Epoch-to-epoch and seed-to-seed variability is of the same order as the differences between several variants, which is why all comparisons in this paper are seed-replicated.

only past poses are observed (the inert-loss condition of our other ablations): minADE 0.6618 ± 0.0366; and (b) *30 fps motion-prior-completed future poses* — the Waymo-3DSkelMo pipeline infills 99% of future timesteps as pseudo-GT, the one setting in which the geodesic and MPJPE terms receive real, non-degenerate gradients: minADE 0.6586 ± 0.0292.

Activating the future-pose losses changes minADE by -0.5% (hierarchical bootstrap p=0.60; vs. the no-pose baseline -0.3%, p=0.82): **not significant**. Giving the rotation-space losses genuine gradients does not reliably improve trajectory accuracy over the inert condition — consistent with our single-run pilot (0.6567 vs. 0.6532), and both conditions carry the same heavy-tailed seeds as every other no-map cell. Whatever benefit pose conditioning provides comes from the past-pose encoder, not the future-pose prediction task.

A.6 Training Dynamics

Figure 3 shows validation minADE over 30 training epochs for the temporal encoding variants, with per-seed bands (five seeds each). Two qualitative observations are stable across seeds: cross-attention without positional encoding tracks the trajectory-only baseline throughout, and the pose-conditioned variants exhibit larger epoch-to-epoch variability than the baseline. The failure mode is a bad basin, not a late-training collapse: both the GRU's failed seeds (plateaus near 0.72–0.83) and the cross-attention+PE failure seed (505) sit high from early in training. At five seeds the xattn+PE band is no longer the narrowest—its three-seed tightness was the sampling artifact discussed in Section 4.3.

A.7 Qualitative Examples

A.8 Original Single-Run Results (Earlier Draft)

Table 6 reports all 19 single-seed runs of the earlier draft, with loss labels corrected per the audit: "geo" terms were inert (zero gradient, Section 4.8), so the functional supervision is what remains after removing them. Runs 009 and 010 are functionally identical; their spread is seed noise. These runs use unrecorded seeds and are superseded by the seed-replicated matrix of Table 7.

A.9 Seed-Replicated Matrix

Table 7 lists every run of the replication matrix (seeds 101/202/303, plus 404/505 for the headline, encoder, map, and pretrain cells). The four audit cells were additionally re-run with the future-pose mask fix (Section 4.8); those masked runs are in the code release.

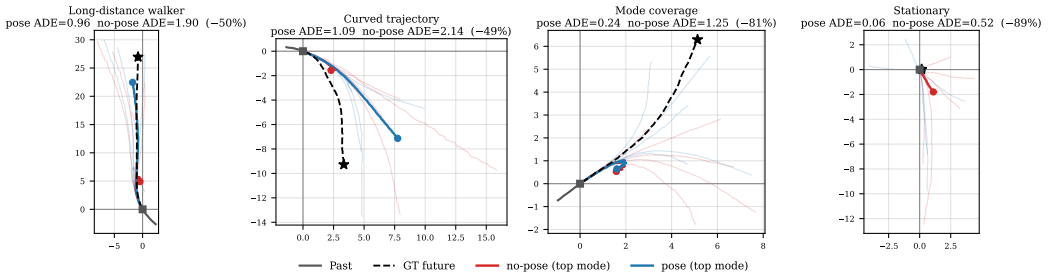


Figure 4: **Qualitative comparison at full-Waymo-pretrained scale.** Four validation scenes. Past in gray; GT future dashed black with ⋆ endpoint. All 6 predicted modes faded; top-scoring mode highlighted (red: no-pose; blue: pose). Per-scene minADE in titles. Scenes selected from the largest per-pedestrian improvements of the pose model, so these illustrate the failure modes pose can fix, not typical magnitudes. *Long-distance:* no-pose stops at 5m; pose follows the 25m walk. *Curved:* no-pose underestimates displacement; pose follows the curve. *Mode coverage:* both top modes underestimate, but pose has a faded mode aligned with GT. *Stationary:* no-pose wrongly predicts motion; pose correctly stays near origin.

Table 6: Earlier-draft single runs (corrected labels). "Functional losses" removes inert terms; w in parentheses. Baseline for Δ is run 004. Runs 001–002 are invalid (zero future trajectory GT).

Run	Draft tag	Encoder	Functional losses / Map / Pretrain	minADE	Δ
001	full pose	GRU	all (1.0)	0.6114	*invalid*
002	no pose	—	traj only	0.7724	*invalid*
003	full	GRU	mpjpe+wta+cls (0.1)	0.6532	−3.2%
004	baseline	—	traj only	0.6745	—
005	full w=0.05	GRU	mpjpe+wta+cls (0.05)	0.6557	−2.8%
006	full w=0.2	GRU	mpjpe+wta+cls (0.2)	0.6542	−3.0%
007	30fps	GRU	mpjpe+geo+wta+cls (0.1), real future	0.6567	−2.6%
008	"mpjpe only"	GRU	mpjpe+wta (0.1)	0.6576	−2.5%
009	"wta only"	GRU	wta (0.1)	0.6467	−4.1%
010	"geo only"	GRU	wta (0.1) [geo inert]	0.6231	−7.6%
011	"geo w=0.05"	GRU	wta (0.05) [geo inert]	0.6786	+0.6%
012	"geo w=0.2"	GRU	wta (0.2) [geo inert]	0.6660	−1.3%
013	cross-attn	XAttn	wta (0.1) [geo inert]	0.6765	+0.3%
014	xattn+PE	XAttn+PE	wta (0.1) [geo inert]	0.6337	−6.0%
015	no-pose+map	—	traj only + map	0.4880	−27.7%
016	pose+map	GRU	wta (0.1) + map [geo inert]	0.4797	−28.9%
017	pretrain	—	traj only + map (full Waymo, 231k peds)	—	*ckpt only*
018	finetune pose	GRU	wta (0.1) + map, pretrained	0.3749	−44.4%
019	finetune no-pose	—	traj + map, pretrained	0.3861	−42.8%

Table 7: Seed-replicated matrix: best-checkpoint minADE per seed (seeds 101/202/303, plus 404/505 for headline cells). All runs 30 epochs, identical protocol; "geo (inert)" marks the zero-gradient geodesic term carried by some configurations.

Cell	Configuration	minADE mean±std	per-seed values
baseline	no pose	0.6604 ± 0.0143	.6550 .6831 .6641 .6541 .6456
geo_pure	GRU, no active aux (geo inert)	0.6545 ± 0.0271	.6958 .6650 .6392 .6259 .6466
wta01	GRU, WTA-L1 + geo (inert)	0.6853 ± 0.0651	.6390 .7848 .6292 .7167 .6570
gmm_only	GRU, WTA-L1 (same config as wta01)	0.7060 ± 0.1067	.8967 .6621 .6621 .6586 .6504
mpjpe	GRU, MPJPE + WTA-L1	0.6723 ± 0.0321	.7243 .6622 .6441 .6801 .6509
full	GRU, MPJPE+WTA+cls (geo inert)	0.6618 ± 0.0366	.6680 .6740 .6302 .6233 .7137
xattn	XAttn no PE, WTA-L1 (+ geo inert)	0.6705 ± 0.0336	.7136 .6408 .6457 .6526 .6996
xattn_pe	XAttn+PE, WTA-L1 (geo inert)	0.6648 ± 0.0381	.6395 .6343 .6374 .7098 .7029
norootorient	XAttn+PE, root-orient zeroed	0.6628 ± 0.0056	.6627 .6666 .6605 .6548 .6694
pose30fps	GRU, full loss, 30 fps future poses	0.6586 ± 0.0292	.6559 .6542 .6338 .6408 .7081
map_nopose	no pose, with map	0.4854 ± 0.0047	.4903 .4839 .4886 .4783 .4860
map_wta01	GRU, WTA-L1, with map	0.4823 ± 0.0058	.4798 .4911 .4853 .4768 .4787
map_xattn_pe	XAttn+PE, WTA-L1, with map	0.4778 ± 0.0041	.4769 .4811 .4795 .4710 .4804
map_norootorient	XAttn+PE, root zeroed, map	0.4800 ± 0.0046	.4766 .4853 .4782
map_agentrot	XAttn+PE, agent-centric pose, map	0.4801 ± 0.0056	.4753 .4862 .4787
ft_nopose	no pose, map, pretrained	0.3745 ± 0.0071	.3815 .3753 .3804 .3706 .3645
ft_wta01	GRU, WTA-L1, map, pretrained	0.3715 ± 0.0042	.3744 .3760 .3651 .3711 .3709
ft_xattn_pe	XAttn+PE, WTA-L1, map, pretrained	0.3718 ± 0.0034	.3710 .3712 .3777 .3694 .3697